

AI Intern Take-Home Assignment: "Dynamic Multimodal-RAG & Fact-Verification Engine"

1. Overview

The goal of this assignment is to evaluate your understanding of building production-ready, asynchronous, and cost-effective AI services. You will build a lightweight FastAPI microservice that dynamically fetches raw text data and visual media from the web based on a user's query, runs a local zero-cost Vector RAG pipeline, and uses a free open-source multimodal vision-language model to cross-verify the media context against the retrieved facts.

2. System Architecture & Core Requirements

Your FastAPI application must expose a single POST endpoint `/analyze` that accepts a JSON payload containing a topic (e.g., `{"topic": "Tesla Model Y"}` or `{"topic": "Human Heart"}`).

The backend workflow must implement the following pipeline layers:

Layer 1: Dynamic Data Fetching (Ingestion)

- Write a module to dynamically query the open/free Wikipedia API.
- Automatically extract the top 3-5 text paragraphs and the URL of the primary/main showcase image associated with that page.

Layer 2: Local Embeddings & Vector Search (RAG)

- Segment the downloaded text content into clean semantic chunks.
- Embed these chunks locally using embedding model.
- Index these embeddings inside a local vector storage layout.

Layer 3: Asynchronous Query & Context Matching

- Create a simple inner semantic search routine: take the topic's core premise, query your vector store, and pull out the top 2 most relevant context chunks.
- Implement your application flows asynchronously to ensure high concurrency when managing background context retrievals.

Layer 4: Multimodal Synthesis & Cross-Verification

- Construct a unified prompt payload containing:
 - The top retrieved text chunks from your local RAG layer.
 - The raw image URL extracted in Layer 1.
- Forward this multimodal context to an open-source VLM/LLM.
- Instruct the model to perform a strict grounded verification check: *Is the fetched image contextually accurate and relevant to the facts stated in the retrieved text chunks?*

3. Expected API Response Format

The `/analyze` endpoint must return a structured JSON response containing strict fact-verification keys. Do not return loose markdown text blocks.

```
```json { "topic": "Human Heart", "execution_metrics": { "time_taken_seconds": 1.42, "chunks_evaluated": 2 }, "retrieved_context": [ "The human heart is a muscular organ...", "It consists of four main chambers..." ], "verification_results": { "image_url": "https://upload.wikimedia.org/.../Heart_diagram.png", "image_is_relevant": true, "confidence_score": 0.95, "synthesis_summary": "The image accurately details the cardiovascular structure, matching the ventricles and atria explained in the retrieved textual documents." } }
```